Anticipating misconceptions and planning how to uncover them

Use this when students repeatedly make the same plausible error. Anticipate the reasoning behind that error so the lesson exposes and repairs it rather than merely announcing the correct answer.

How to do it well

1 Start from the intended idea. Write the accurate rule, relationship or distinction students need to understand.

2 Predict the wrong model. Describe what a student might reasonably believe and why that would produce the error.

3 Design the diagnostic prompt. Use a choice, comparison, explanation or short task that distinguishes the correct and incorrect reasoning.

4 Prepare the repair. Select an example and non-example, model or demonstration that addresses the precise difference.

5 Plan the fresh check. Change the context enough to require thinking while retaining the concept you are checking.

What this sounds like

“Someone says heavier objects always fall faster. Which part of that claim needs testing? Explain what evidence would settle it.”

Classroom example. In Geography, a teacher anticipates the belief that weather and climate mean the same thing. Students classify statements about a single day and long-term patterns. Their explanations reveal whether the distinction is secure. The teacher contrasts the timescales explicitly, then gives new statements so pupils apply the repaired distinction.

Top tips that make the technique work

Use real errors. Student work is a better source of plausible misconceptions than invented distractors with no likely rationale.

Separate slips from beliefs. Ask how the answer was reached before assuming a conceptual misconception.

Plan a near-miss. A convincing wrong example can make the critical distinction visible.

Keep the correction focused. Repair the faulty rule or decision rather than explaining every part of the topic again.

Recheck with changed surface features. Students need to apply the repaired idea, not repeat the exact answer they just heard.

Do and avoid

Do	Avoid
Name the misconception each planned answer would reveal.	Calling every incorrect answer a misconception.
Prepare a contrasting example and follow-up question.	Revealing the solution before finding out students’ reasoning.

Further reading

Practitioner guidance; teacher language above is an illustrative BSCS adaptation.

Barton - Diagnostic questions